Data-Driven Investigation of Culinary Patterns in CulinaryDB

Exploratory Data Analysis Report for Research Presentation

Inspired by Singh & Bagler (2018) and Sharma et al. (2020) hierarchical clustering paper

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Executive Summary

This report presents a beginner-friendly yet research-aligned exploratory data analysis (EDA) of CulinaryDB, a curated repository of 45,772 traditional recipes spanning 26 world cuisines. The work connects to prior research from Dr. Ganesh Bagler's lab on culinary pattern mining, ingredient popularity, food-pairing behavior, and hierarchical clustering of cuisines.

Background of prior research

- Singh & Bagler (2018) analyzed 45,772 recipes across 22+ regions and showed that ingredient popularity, recipe size, and rank-frequency distributions follow consistent non-random patterns.
- Sharma et al. (2020, arXiv:2004.12283) extended this direction by mining frequent ingredient/process patterns and building phylogenetic-style hierarchical trees to study geographical and historical relatedness of cuisines.
- Jain et al. (2015) demonstrated that Indian regional cuisines exhibit negative food pairing driven largely by spices.

Key dataset highlights

- Total recipes analyzed: 45,772
- Number of cuisines: 26
- Unique ingredient entities: 695
- Global average recipe size: 9.01 ingredients
- Most frequent ingredient overall: salt (18,745 recipes)
- Largest cuisine by recipe count: USA (16,118 recipes)
- Largest ingredient vocabulary: USA (595 unique ingredients)

What we reproduce in this report

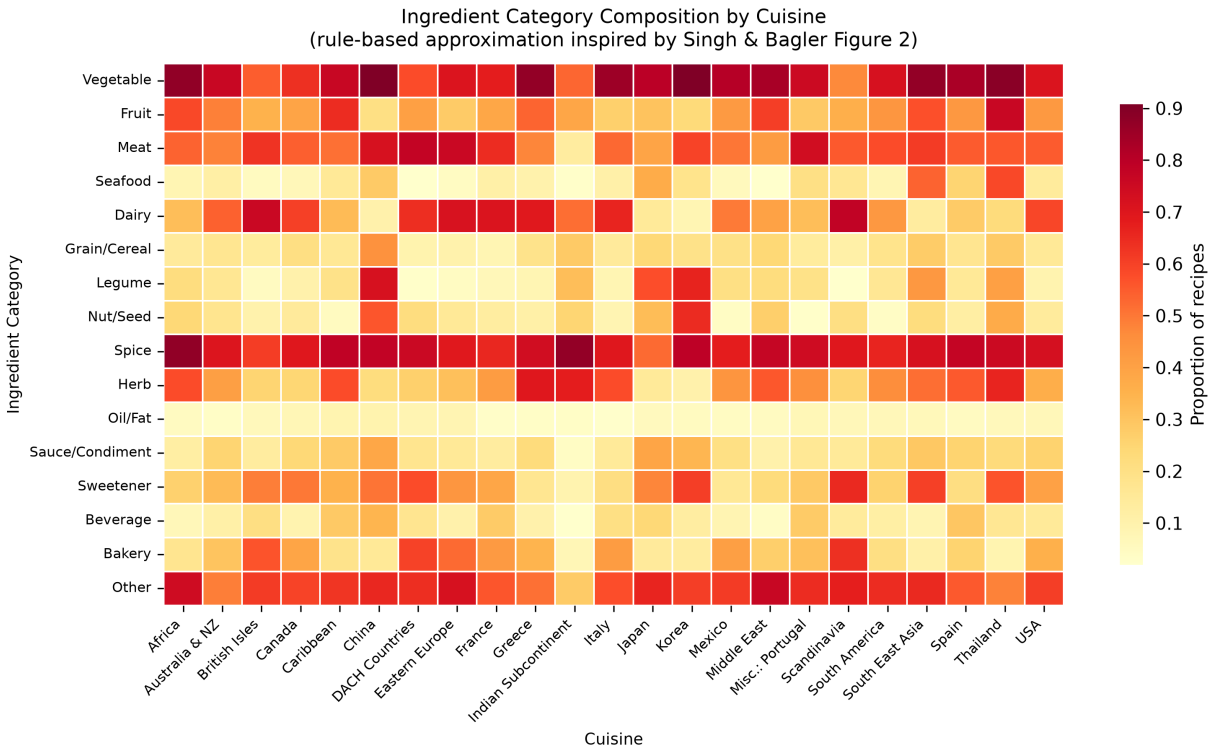
- Category composition heatmap across cuisines (Singh & Bagler style)
- Recipe size distribution curves and box plots
- Power-law / log-log rank-frequency plot of ingredient popularity
- Bar charts with standard deviation error bars for average recipe size
- Phylogenetic-style hierarchical clustering trees (Sharma et al. paper)
- Authenticity-based cuisine similarity dendrogram and heatmap
- Basic EDA: recipe counts, unique ingredients, top ingredients, source distribution

Scope and limitations

CulinaryDB provides recipes and normalized ingredients but not cooking processes, utensils, or flavor-molecule profiles. Therefore, analyses requiring FlavorDB (food-pairing Z-scores, molecule overlap) are documented as out of scope rather than approximated with synthetic values. Pattern mining here uses ingredient co-occurrence only, which is a simplified adaptation of the full RecipeDB pipeline in the hierarchical clustering paper.

1. Ingredient Category Composition

Figure 1 - Ingredient Category Composition by Cuisine



How this figure was created

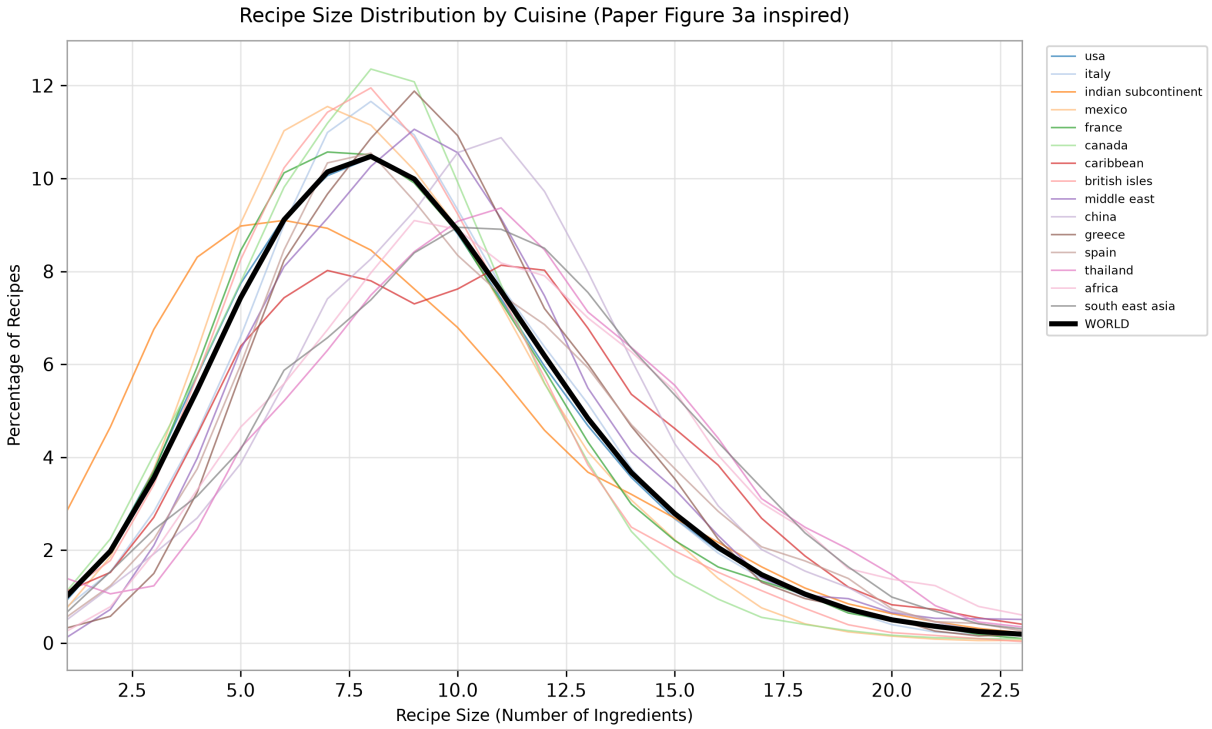
Each ingredient was mapped to one of 16 rule-based categories using keyword matching and CulinaryDB category labels. For each cuisine, we computed the proportion of recipes containing at least one ingredient from each category. The matrix was visualized using a seaborn heatmap (YlOrRd colormap).

Key insights and interpretation

- Spices, vegetables, and sauces/condiments appear in a large fraction of recipes across most cuisines.
- Meat, seafood, dairy, and legume proportions vary strongly by cuisine and reveal regional dietary signatures.
- Bright horizontal bands indicate categories that are nearly universal (e.g., spices and vegetables).
- Normalization by recipe count allows fair comparison despite dataset imbalance across cuisines.

2. Recipe Size Analysis

Figure 2 - Recipe Size Distribution by Cuisine



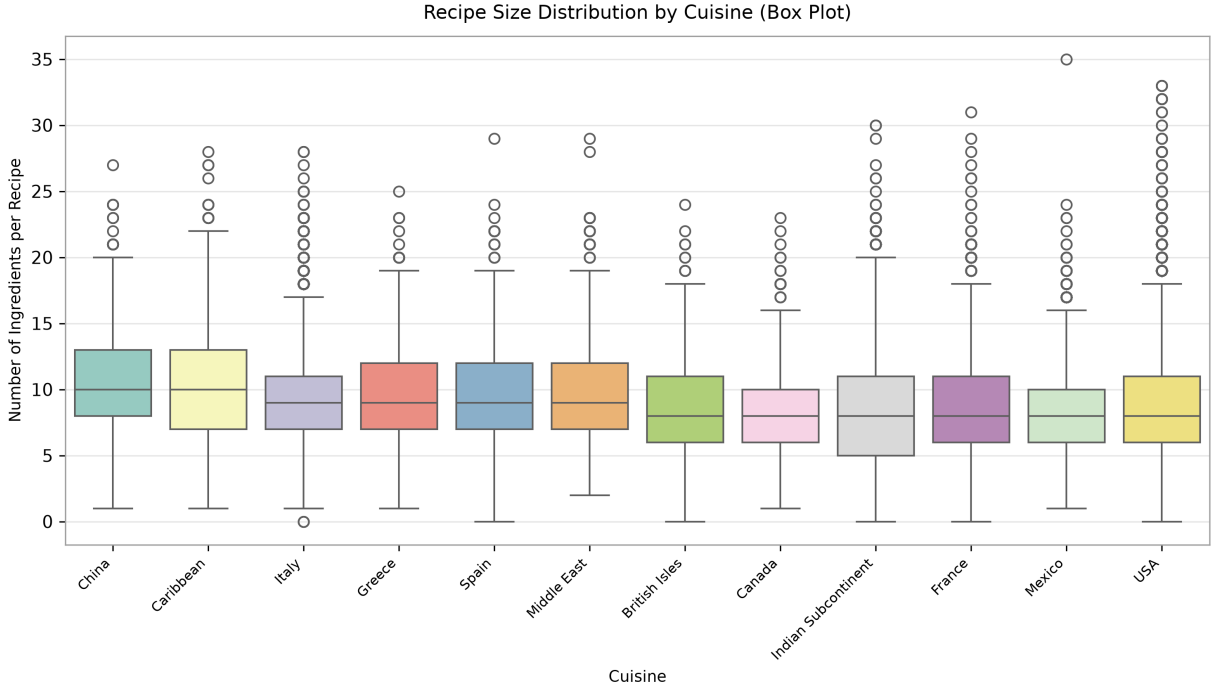
How this figure was created

Recipe size was defined as the count of distinct ingredient entities per recipe. For each cuisine and for WORLD, we computed the percentage of recipes at each ingredient count (1 to 23) and plotted smoothed line curves.

Key insights and interpretation

- The global mean recipe size is about 9.01 ingredients, close to the ~9 reported in prior papers.
- Most cuisines show a unimodal distribution: frequency rises to a peak then declines.
- Thai, African, and South East Asian cuisines tend toward larger recipes on average.
- The thick WORLD curve summarizes the shared central tendency across all cuisines.

Figure 3 - Recipe Size Box Plot by Cuisine



How this figure was created

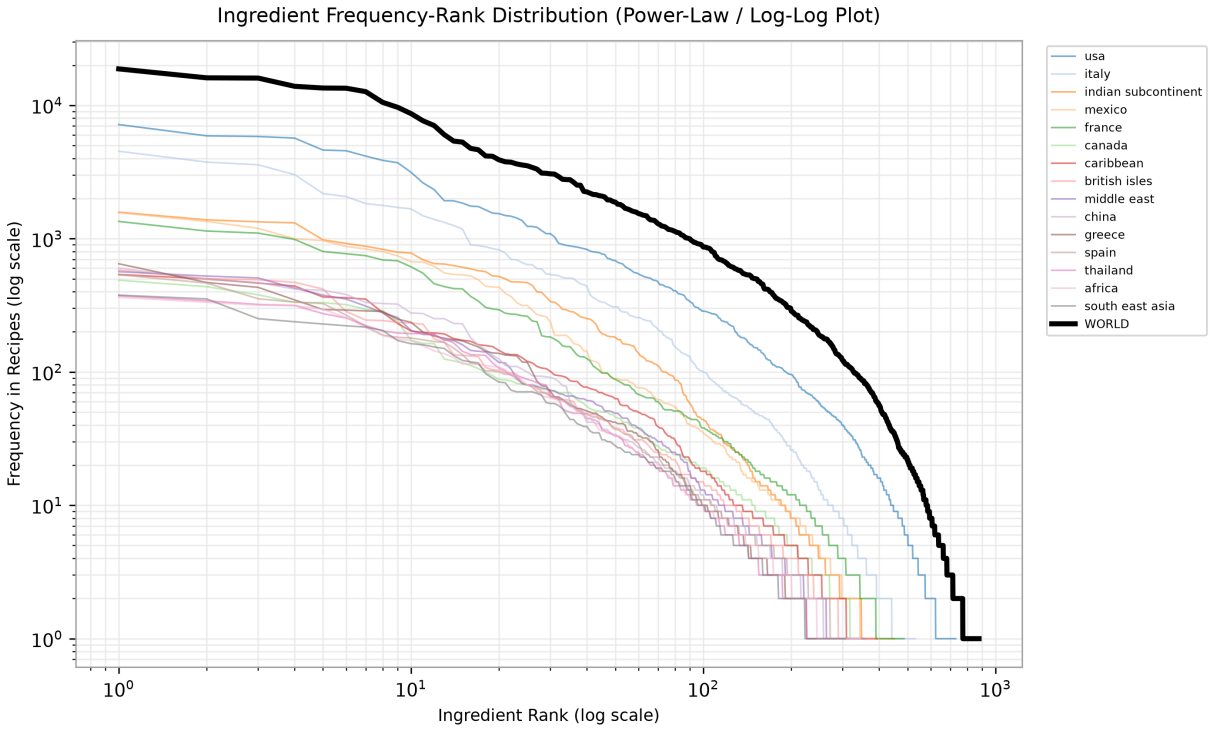
A box plot was created for the top 12 cuisines by recipe count, showing median, quartiles, and outliers of recipe size (ingredient count per recipe).

Key insights and interpretation

- Box plots reveal spread and skew beyond what averages alone can show.
- Wide boxes indicate high variability in recipe complexity within a cuisine.
- Outlier points above the whiskers represent unusually ingredient-heavy recipes.
- Medians may differ from means when distributions are skewed.

3. Ingredient Popularity and Power-Law Behavior

Figure 4 - Ingredient Frequency-Rank Distribution (Log-Log / Power-Law Plot)



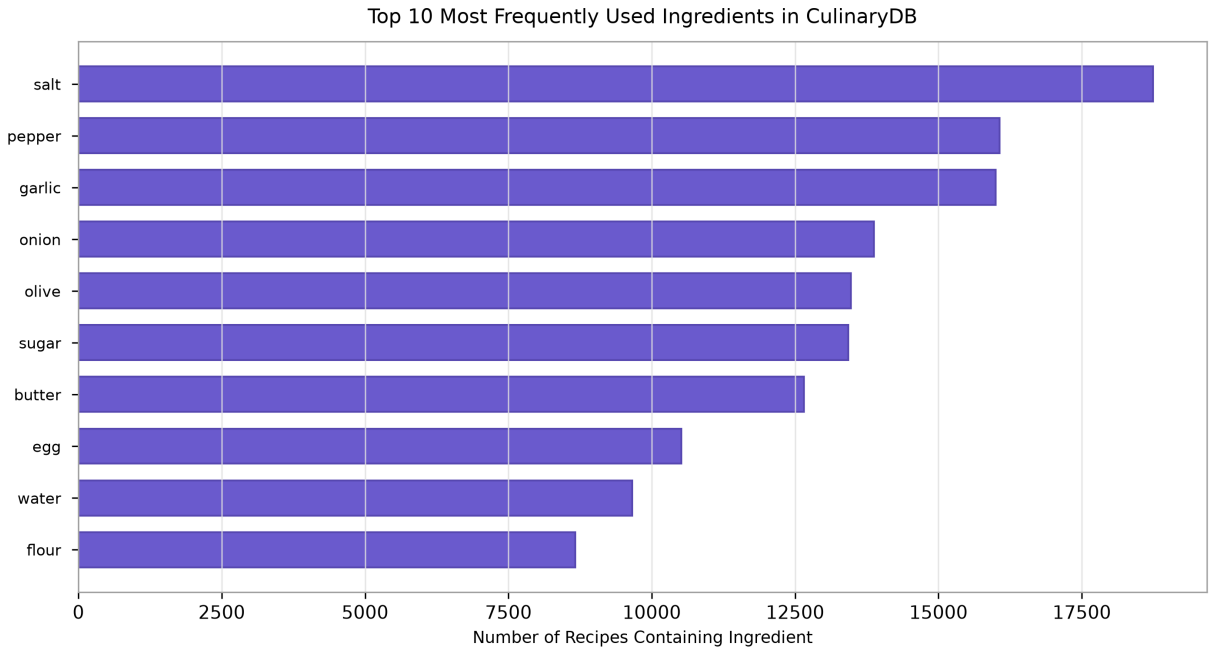
How this figure was created

For each cuisine and for WORLD, ingredients were ranked by the number of recipes containing them. Rank (x-axis) and frequency (y-axis) were plotted on logarithmic scales, reproducing the classic head-heavy pattern reported in Singh & Bagler Figure 3b.

Key insights and interpretation

- Every cuisine shows a steep drop from rank 1 to higher ranks -- a hallmark of power-law-like behavior.
- A small set of staple ingredients (salt, pepper, onion, garlic, oil) dominate most cuisines.
- The long tail represents many rarely used specialty ingredients.
- Different cuisines differ in how quickly frequency decays with rank, reflecting local ingredient diversity.

Figure 5 - Top 10 Most Frequent Ingredients Overall



How this figure was created

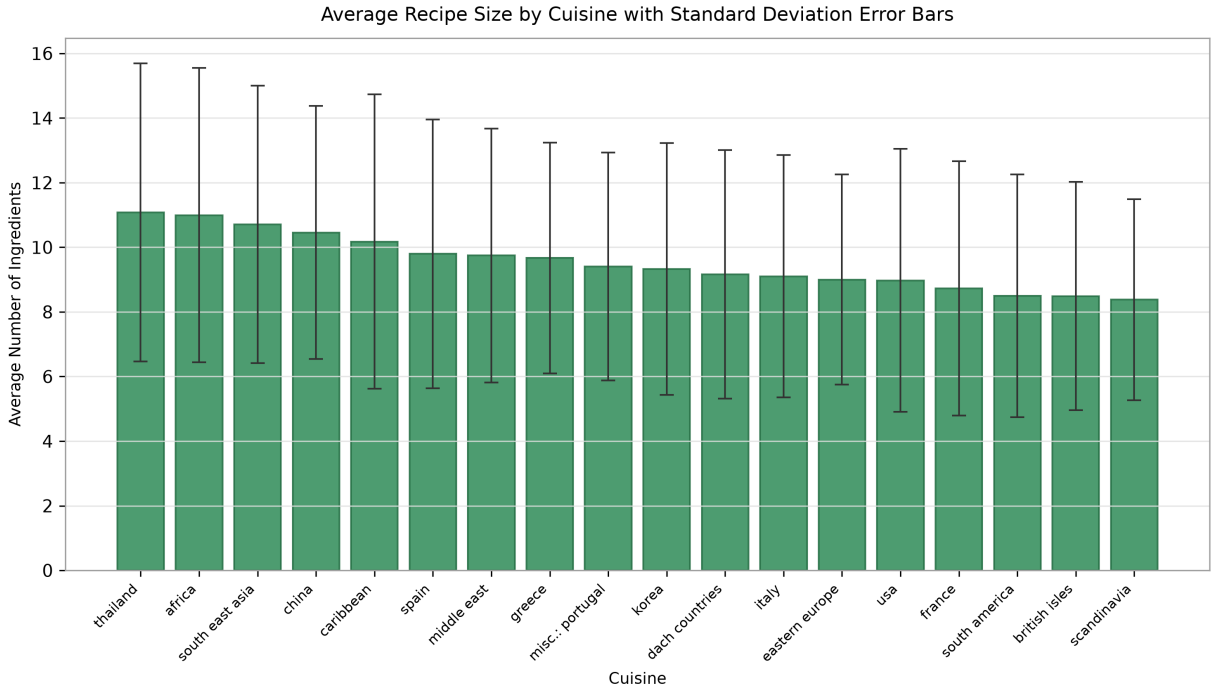
We counted how many distinct recipes contain each ingredient (one count per recipe per ingredient) and plotted the top 10 as a horizontal bar chart.

Key insights and interpretation

- The most common ingredient is salt (18,745 recipes).
- Basic pantry staples dominate the global top ranks.
- Frequency counts recipes, not total alias occurrences in the database.
- These results reflect recipe-writing conventions in the CulinaryDB collection.

4. Cuisine Comparison with Error Bars

Figure 6 - Average Recipe Size with Standard Deviation Error Bars



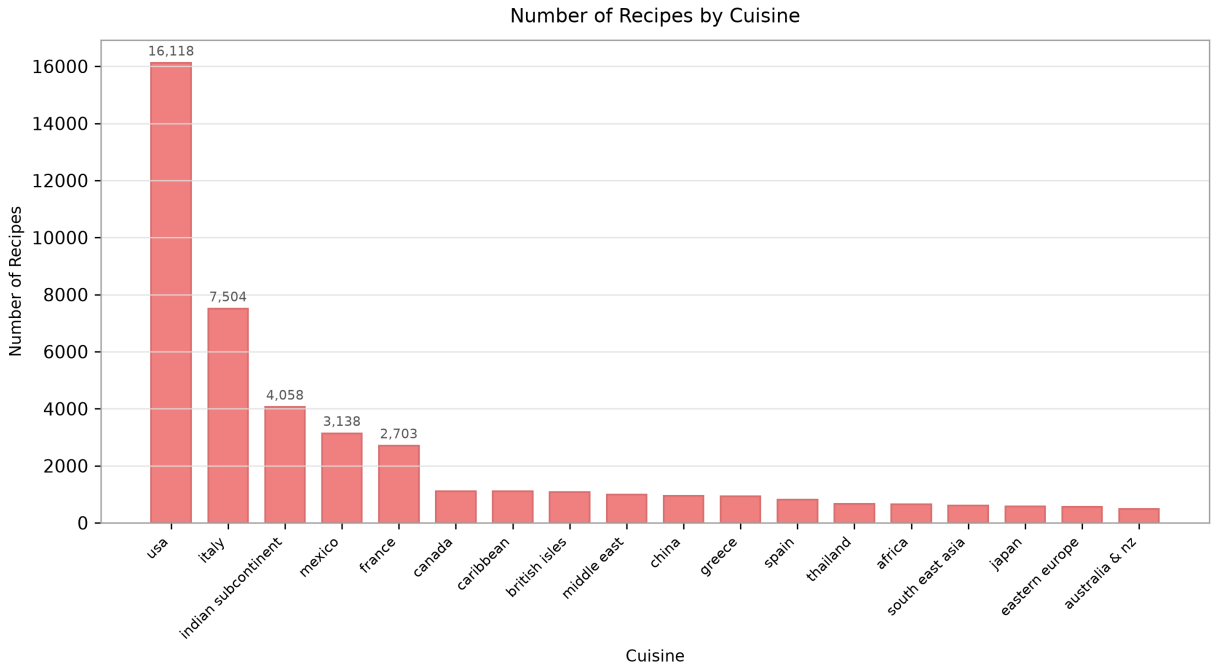
How this figure was created

For cuisines with at least 100 recipes, we computed mean recipe size and standard deviation. A bar chart with error bars shows both central tendency and spread.

Key insights and interpretation

- Highest average recipe size among major cuisines: Thailand (11.08 ingredients).
- Long error bars indicate high within-cuisine variability in recipe complexity.
- Short error bars suggest more uniform recipe sizes within that cuisine.
- Error bars represent standard deviation, not standard error of the mean.

Figure 7 - Number of Recipes by Cuisine



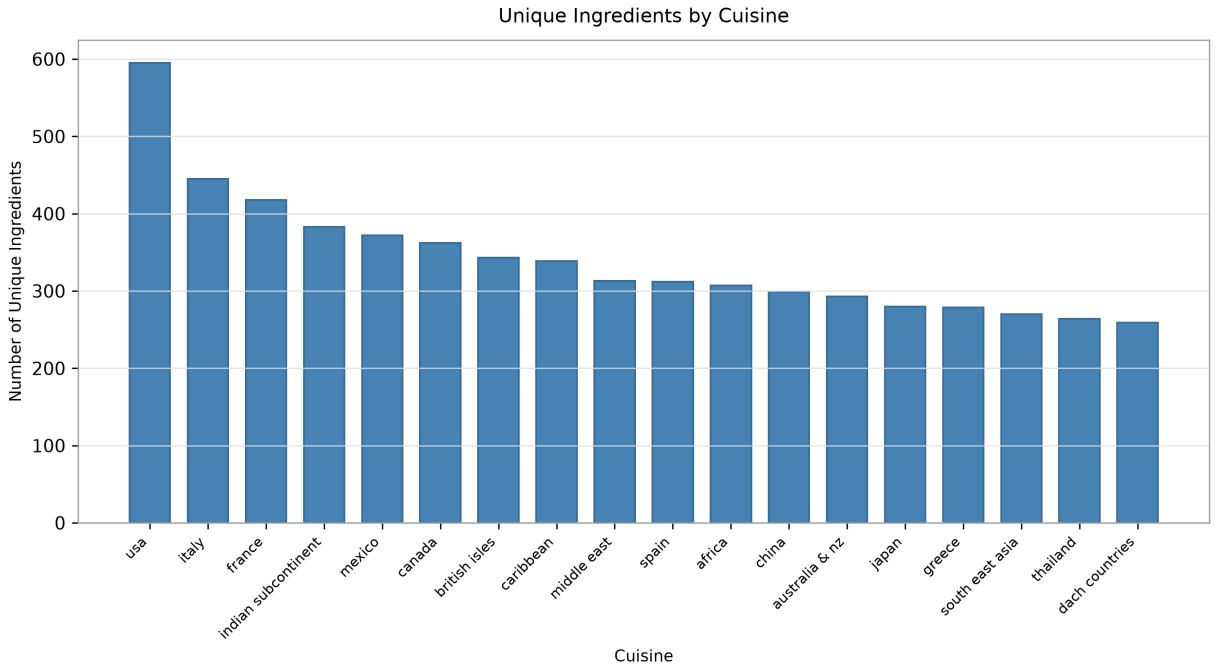
How this figure was created

Simple value counts of recipes per cuisine, sorted descending and plotted as a bar chart.

Key insights and interpretation

- USA dominates the dataset with 16,118 recipes.
- Italy and Indian Subcontinent are the next largest cohorts.
- The dataset is heavily imbalanced across cuisine classes.
- Imbalance is noted for context; no classification model was trained.

Figure 8 - Unique Ingredients by Cuisine



How this figure was created

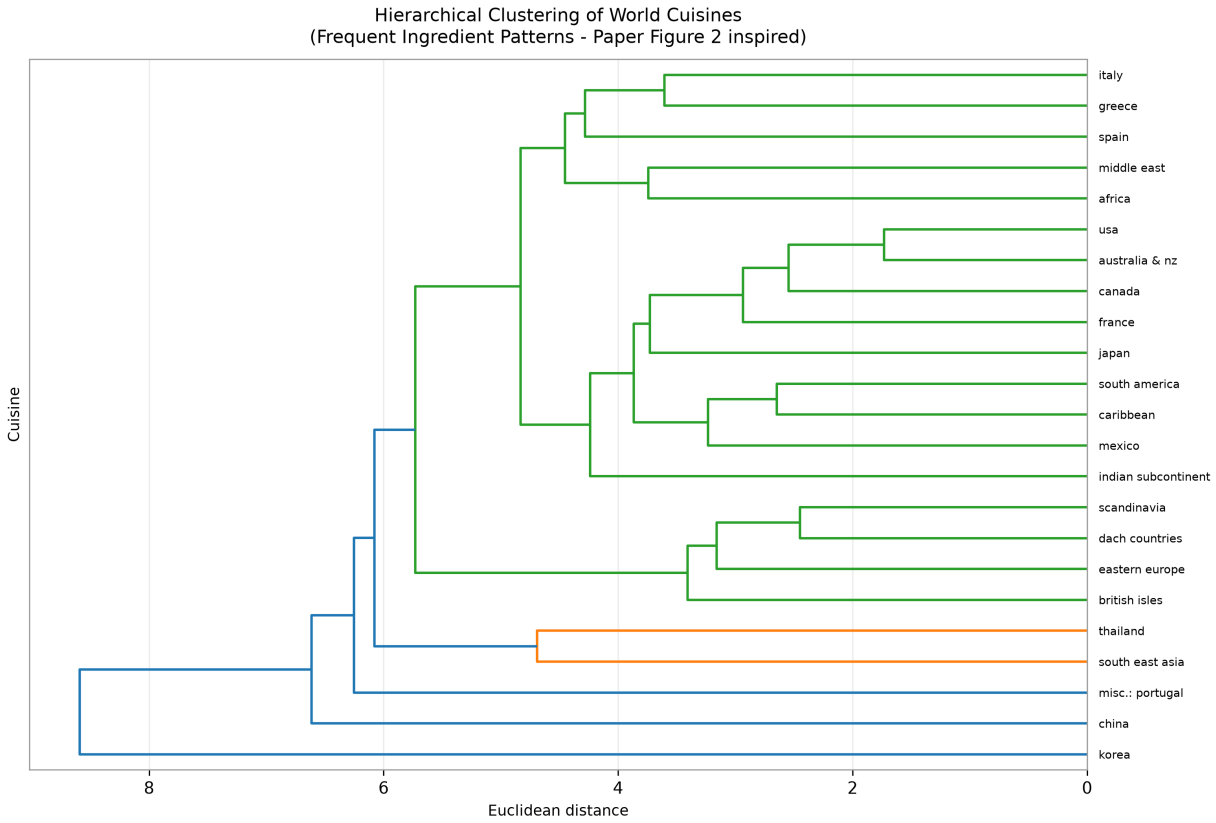
For each cuisine, we counted distinct ingredient entity IDs across all recipes and plotted the top 18 cuisines.

Key insights and interpretation

- USA has the largest ingredient vocabulary (595 entities).
- Large recipe corpora tend to accumulate more unique ingredients in the database.
- Smaller cuisines can still show substantial diversity relative to their recipe count.
- This measures breadth in the dataset, not real-world culinary completeness.

5. Phylogenetic / Hierarchical Clustering of Cuisines

Figure 9 - Hierarchical Tree using Euclidean Distance on Frequent Patterns



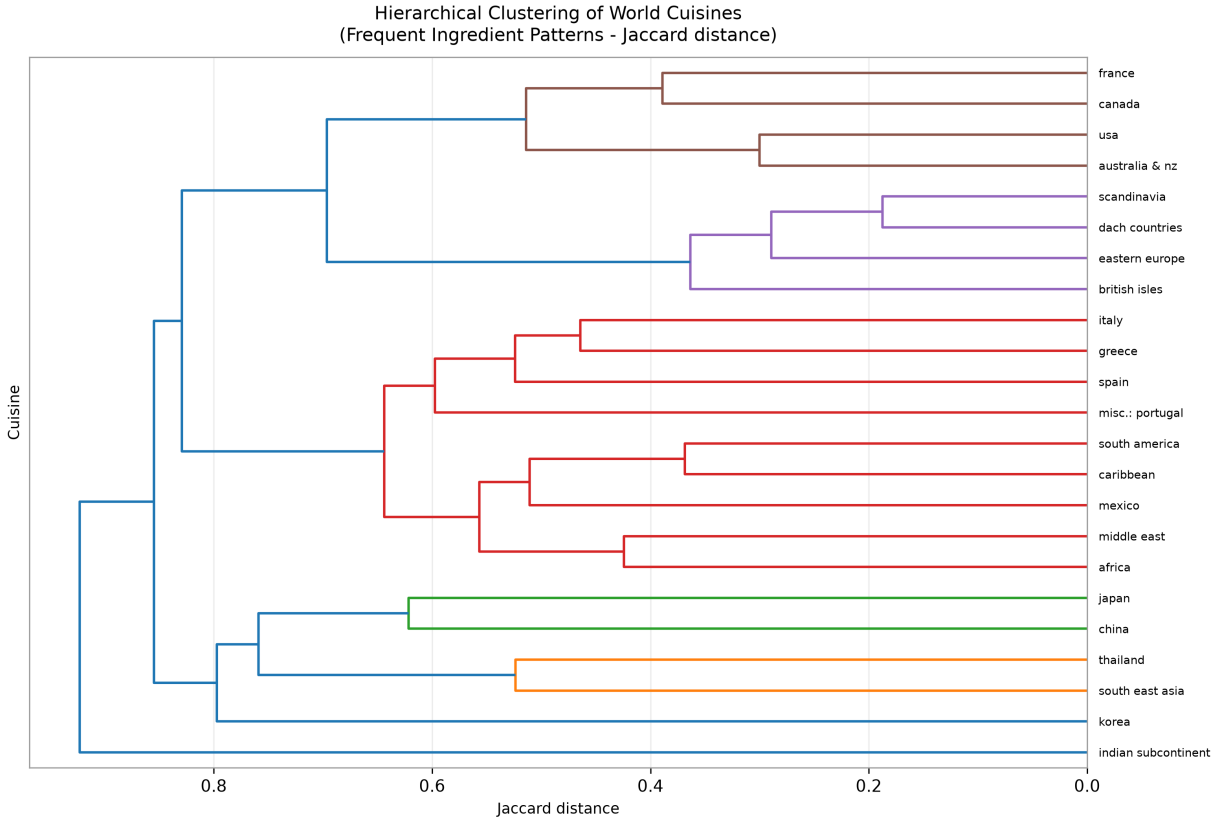
How this figure was created

We mined frequent ingredient co-occurrence patterns (support >= 20%) within each cuisine, encoded them as binary feature vectors, computed pairwise Euclidean distances, and applied average-linkage hierarchical clustering to produce a dendrogram -- analogous to Sharma et al. Figure 2.

Key insights and interpretation

- Cuisines that merge at lower linkage height are more similar in their frequent ingredient patterns.
- European and American cuisines often cluster together, reflecting shared pantry staples.
- Asian cuisines (Japanese, Korean, China, Thai) form a distinct branch in many cases.
- This is an ingredient-only adaptation; the original paper also used processes and utensils.

Figure 10 - Hierarchical Tree using Jaccard Distance



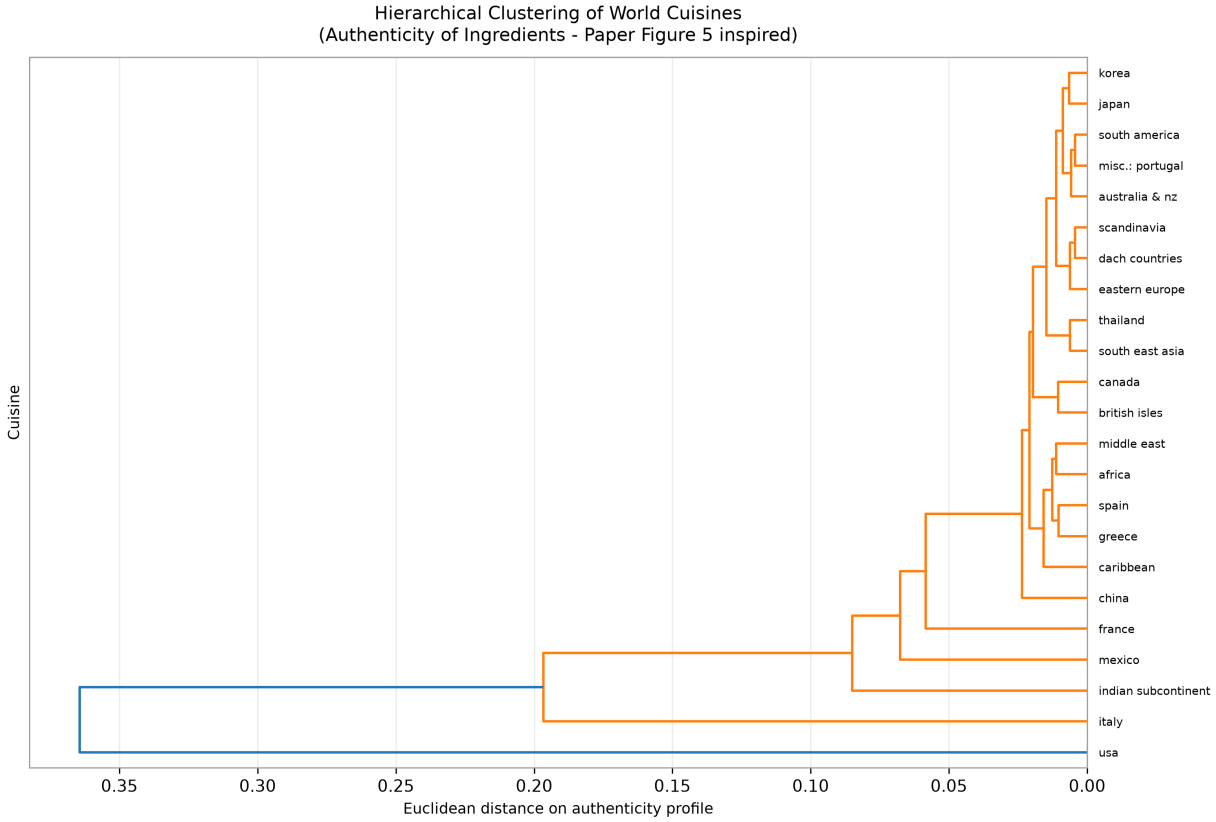
How this figure was created

The same frequent-pattern vectors were clustered using Jaccard distance, which measures dissimilarity based on shared vs. total patterns -- better suited for binary presence/absence data.

Key insights and interpretation

- Jaccard distance emphasizes pattern overlap rather than vector magnitude.
- Cluster structure may differ from Euclidean clustering for the same data.
- Geographically distant cuisines can cluster together if they share ingredient patterns.
- Historical connections (e.g., colonial influences) may appear as unexpected groupings.

Figure 11 - Authenticity-Based Phylogenetic Tree



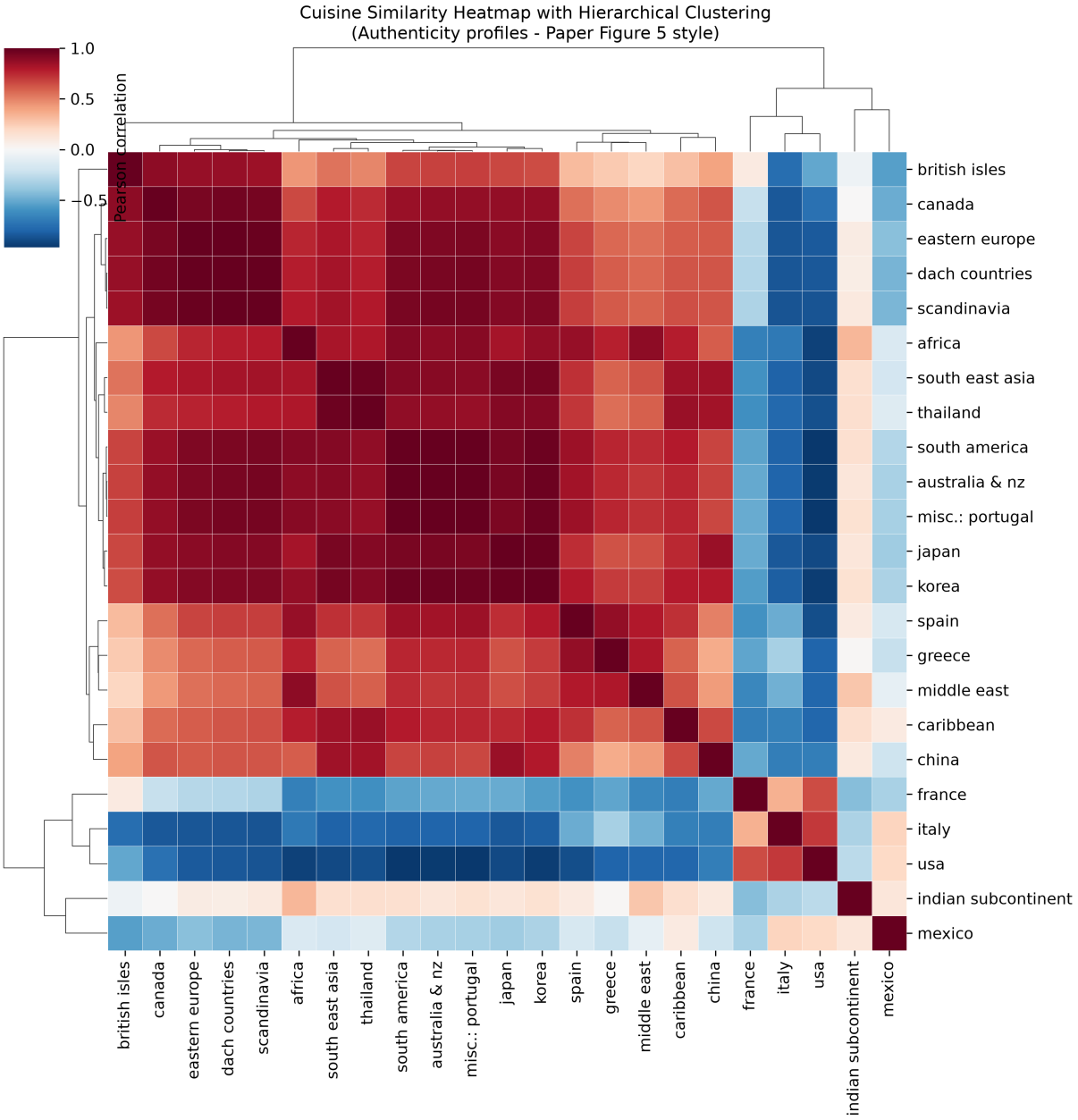
How this figure was created

Following Sharma et al. Section V-B, we computed ingredient prevalence per cuisine, subtracted the cross-cuisine mean prevalence to obtain an authenticity score, and clustered cuisines using Euclidean distance on the top 80 most distinctive ingredient features.

Key insights and interpretation

- Authenticity captures both over-represented and under-represented ingredients vs. global average.
- Indian Subcontinent may cluster closer to spice-heavy regions than to geographically neighboring cuisines.
- Canadian and French cuisines can appear closer due to shared historical culinary influence.
- This tree reflects ingredient identity, not geographic distance alone.

Figure 12 - Cuisine Similarity Heatmap with Dendrograms



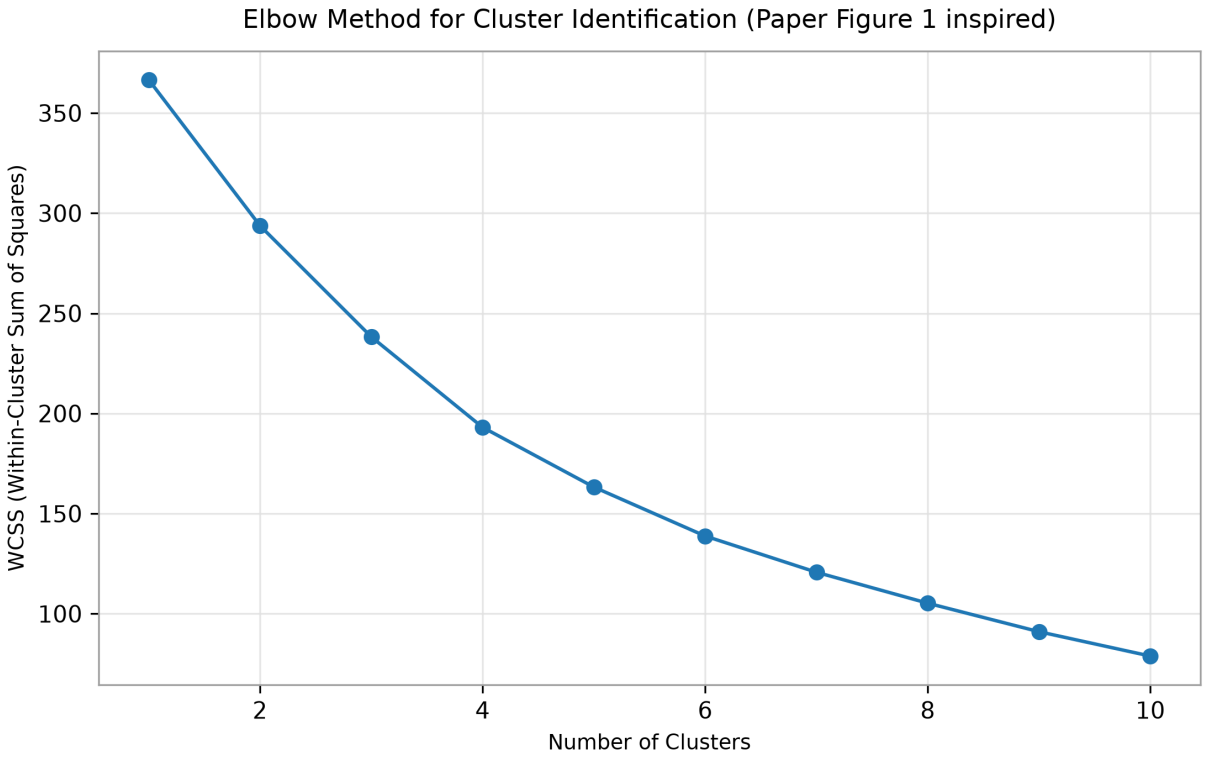
How this figure was created

Pearson correlation of authenticity profiles was computed between all cuisine pairs. A clustermap with row/column dendrograms visualizes both pairwise similarity and hierarchical structure.

Key insights and interpretation

- Red blocks along the diagonal indicate groups of mutually similar cuisines.
- Blue off-diagonal regions highlight cuisines with contrasting ingredient profiles.
- Side dendrograms reorder cuisines to place similar ones adjacent.
- This combines correlation analysis with hierarchical clustering in one view.

Figure 13 - Elbow Method for Cluster Count



How this figure was created

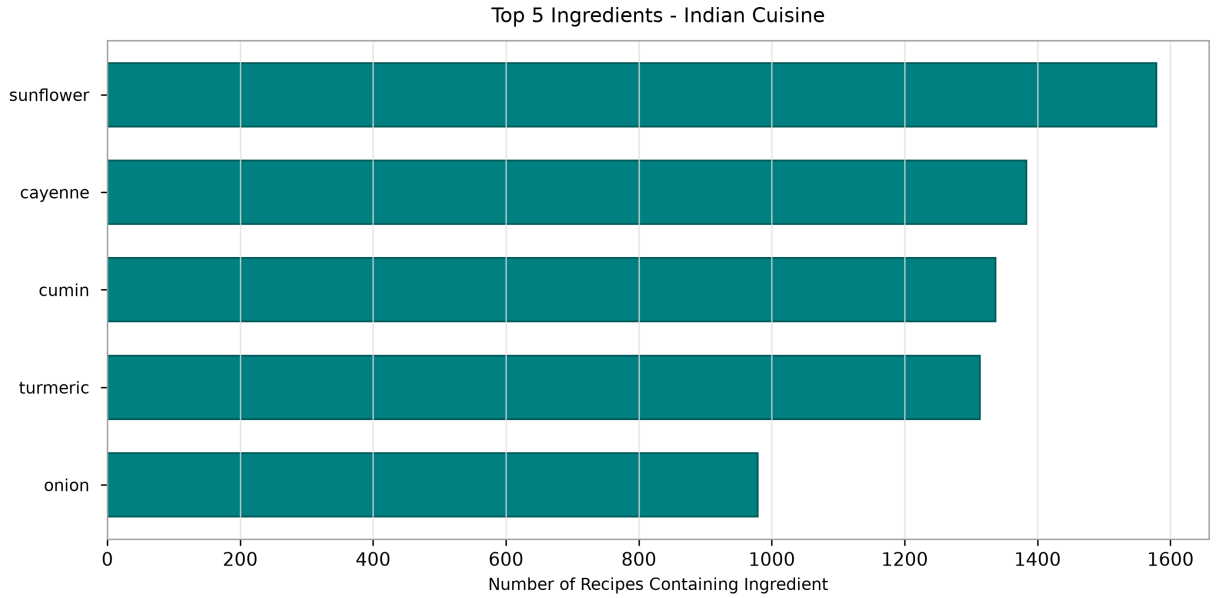
K-means clustering was applied to the frequent-pattern feature vectors for k = 1 to 10. Within-Cluster Sum of Squares (WCSS) was plotted against k, reproducing the elbow analysis from Sharma et al. Figure 1.

Key insights and interpretation

- WCSS decreases monotonically as k increases.
- No sharp elbow is visible, matching the original paper's observation.
- This supports using hierarchical clustering over K-means for this categorical data.
- The lack of a clear elbow suggests cuisine structure is continuous rather than sharply partitioned.

6. Regional Ingredient Signatures

Figure 14 - Top 5 Ingredients in Indian Cuisine



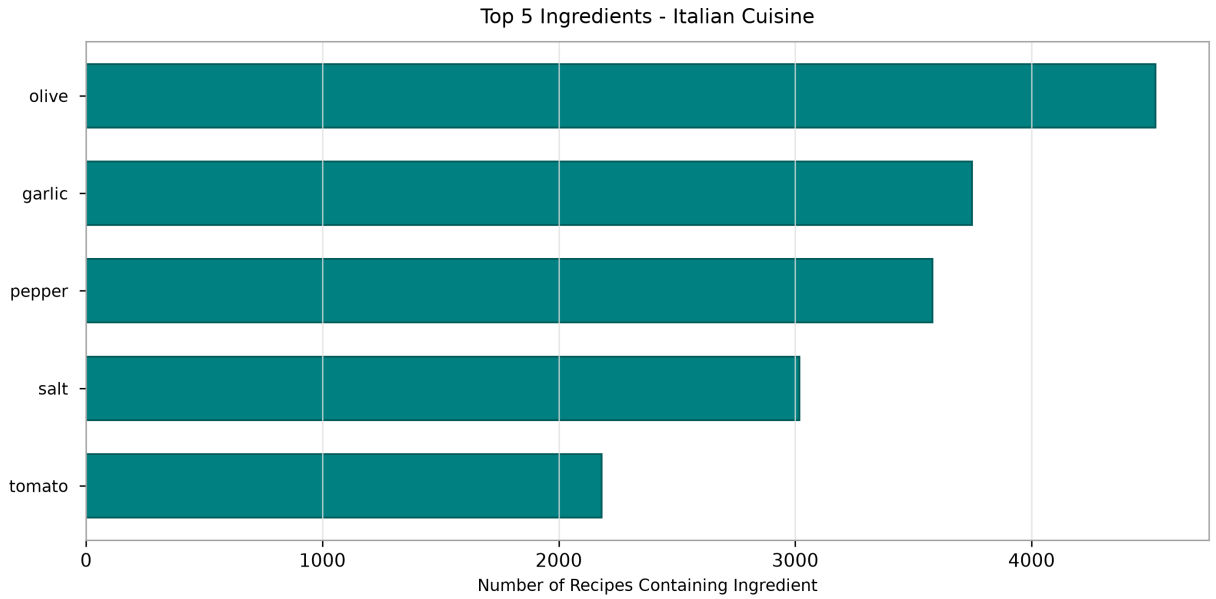
How this figure was created

Filtered recipes to Indian Subcontinent, counted recipe-level ingredient frequency, selected top 5.

Key insights and interpretation

- Indian recipes are dominated by spices and aromatics (cumin, turmeric, cayenne, sunflower oil).
- Regional top ingredients differ substantially from the global top 10.
- Spice-heavy profiles align with prior Bagler lab findings on Indian food pairing.
- Results describe this dataset subset, not every traditional Indian recipe.

Figure 15 - Top 5 Ingredients in Italian Cuisine



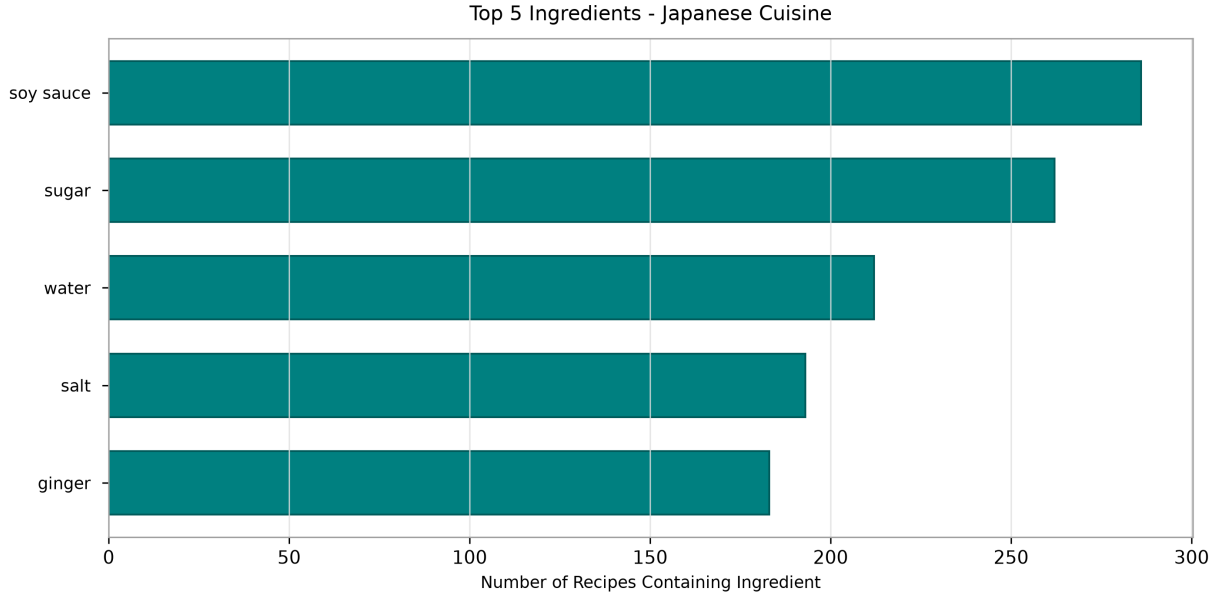
How this figure was created

Same method applied to Italian recipes.

Key insights and interpretation

- Olive oil, garlic, black pepper, and tomato form the Mediterranean flavor foundation.
- Italian profiles emphasize dairy, herbs, and olive oil over spice intensity.
- Overlap with global staples exists but ranking order differs.
- Parmesan and pasta-related ingredients are prominent in the full ranking beyond top 5.

Figure 16 - Top 5 Ingredients in Japanese Cuisine



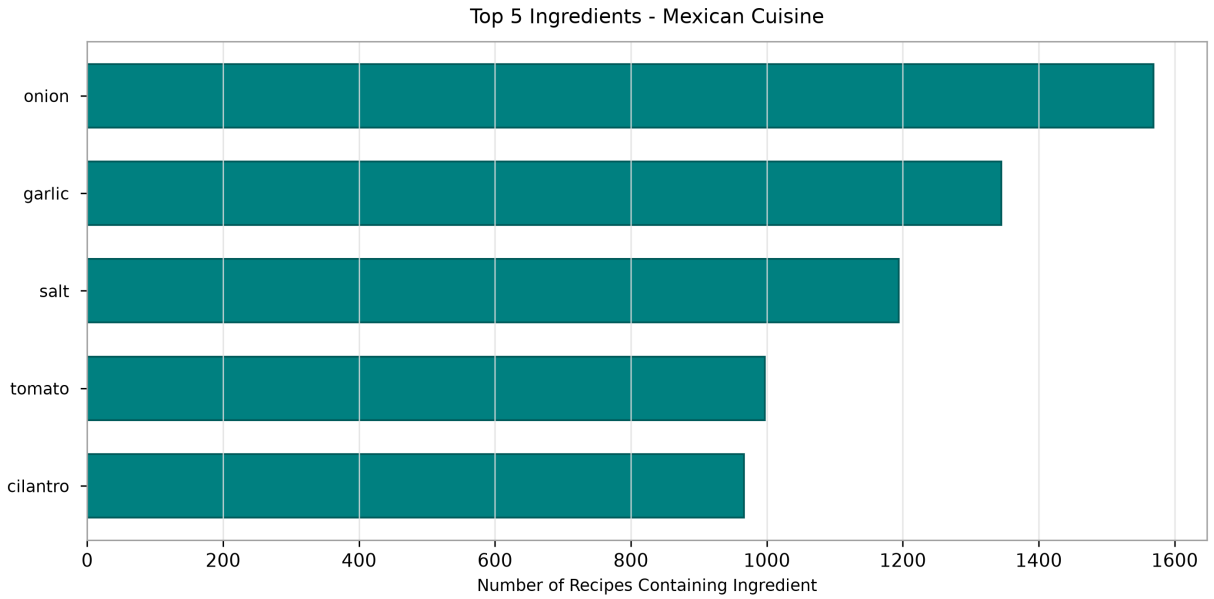
How this figure was created

Same method applied to Japanese recipes.

Key insights and interpretation

- Soy sauce, sugar, water, and ginger dominate Japanese seasoning profiles.
- Umami-rich condiments define the cuisine's character in this dataset.
- Japanese ingredient profiles are distinct from both Indian and Italian signatures.
- Smaller recipe count (580) means rankings are more sensitive to individual recipes.

Figure 17 - Top 5 Ingredients in Mexican Cuisine



How this figure was created

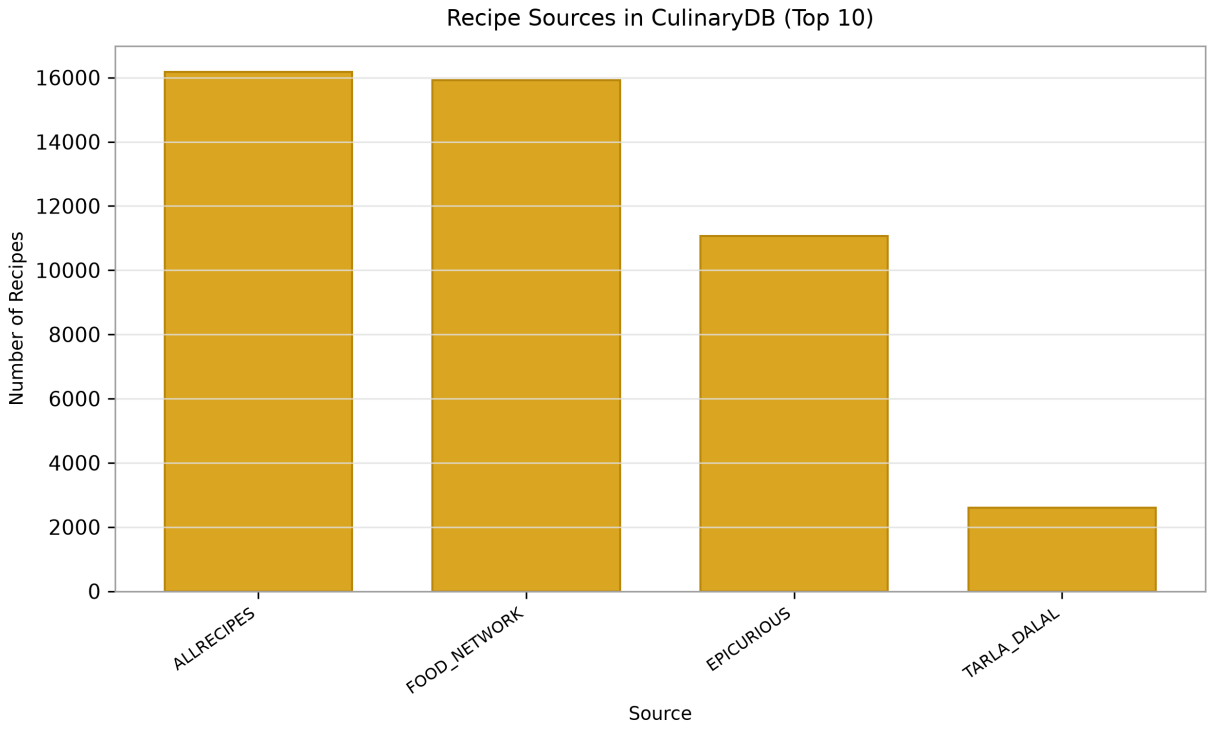
Same method applied to Mexican recipes.

Key insights and interpretation

- Onion, garlic, salt, tomato, and cilantro characterize Mexican flavor foundations.
- Overlap with Mediterranean staples (onion, garlic, tomato) reflects shared plant-based bases.
- Cilantro and chili-related ingredients distinguish Mexican from European profiles.
- Mexican cuisine shows moderate recipe count (3,138) providing stable rankings.

7. Dataset Provenance

Figure 18 - Recipe Sources in CulinaryDB



How this figure was created

Value counts of the Source column in recipe metadata, showing which repositories contributed recipes.

Key insights and interpretation

- Multiple online recipe repositories contribute to CulinaryDB.
- Source imbalance may influence ingredient naming conventions and recipe style.
- TarlaDalal contributes heavily to Indian Subcontinent recipes.
- Understanding sources helps interpret potential collection biases.

Conclusions and Future Work

Key findings

- CulinaryDB contains 45,772 recipes across 26 cuisines with an average recipe size of 9.01 ingredients.
- Ingredient popularity follows a head-heavy rank-frequency pattern consistent with prior published work.
- Cuisine-specific category compositions and top-ingredient profiles reveal distinct regional signatures.
- Hierarchical clustering trees group cuisines by ingredient patterns, sometimes reflecting geography and sometimes history.
- Standard deviation error bars show substantial within-cuisine variability in recipe complexity.

Analyses not reproduced (require external data)

- Food-pairing Z-scores and flavor-molecule overlap (requires FlavorDB).
- Full FP-Growth on processes and utensils (not available in CulinaryDB CSV release).
- Geographic-distance validation dendrogram (requires external geographic coordinates).

References

- Singh, N., & Bagler, G. (2018). Data-driven investigations of culinary patterns in traditional recipes across the world. ICDEW.
- Sharma, T., et al. (2020). Hierarchical Clustering of World Cuisines. arXiv:2004.12283.
- Jain, A., Rakhi, N., & Bagler, G. (2015). Analysis of Food Pairing in Regional Cuisines of India. PLoS ONE.
- CulinaryDB: 01_Recipe_Details.csv, 02_Ingredients.csv, 03_Compound_Ingredients.csv, 04_Recipe-Ingredients_Aliases.csv.